Concept of Operations (ConOps) — Claude apps gateway (AWS GovCloud)

Companion to the security-review package (`architecture.md`, `network-access-controls.md`, `security-assessment-2026-07.md`) for the RMF/FedRAMP ATO submission. Where those documents describe *what the system is built from*, this one describes *how it is operated and used*: who the users are, what they do, how the system behaves in normal and degraded modes, and what the operating organization must provide. It is written for ISSO/AO reviewers who have not read the repository.

This deployment is a **client-configurable template**, not a single fielded system: every organization-specific value (gateway FQDN, Okta issuer, email domains, model IDs, network CIDRs, account IDs) is a CloudFormation parameter or a `scripts/deploy.env` variable. Placeholders in this document (`claude-gateway.example.com`, `grafana-admins`) stand in for those values.

Where the architecture is already documented, this ConOps **references** the diagrams and inventories rather than restating them. When code and prose disagree, the CloudFormation templates in `cloudformation/` are authoritative.

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1. Purpose and scope

1.1 Purpose

The system is a **self-hosted gateway that lets developers use Claude Code (the Anthropic coding agent) entirely inside the organization's AWS GovCloud boundary.** Developer requests never leave to a public Anthropic endpoint; model inference is served from **Amazon Bedrock** within GovCloud (`us-gov-west-1`), authenticated by the organization's own **Okta** identity provider, and recorded in the organization's own audit and observability stores. The gateway exists so that an approved-model, identity-attributed, fully in-boundary path to Claude can be offered to developers on managed Windows laptops without a public internet dependency on the AI provider.

1.2 Scope — what the system is

- An **internal Application Load Balancer + ECS Fargate gateway** service that terminates developer TLS, authenticates users via Okta OIDC, and proxies inference to Bedrock (`cloudformation/02-gateway.yaml`).
- An **RDS PostgreSQL** store for session and spend state (`cloudformation/01-database.yaml`).
- An **offline Windows client rollout**: a mirrored, integrity-verified `claude` binary and a **no-admin, user-scope installer** — it writes the binary and a user-settings `env` block (telemetry tags, update lockdown, CA trust) and needs no elevation. The gateway **login** requires a managed setting (`forceLoginMethod: "gateway"` + `forceLoginGatewayUrl`) that Claude Code honors only from a managed source, delivered by a separate GPO/MDM channel (`scripts/mirror/mirror-claude-release.sh`, `client/Install-ClaudeCode.ps1`; `client-config.md`, `ad-request-email.md`).

- An **optional usage/cost observability stack**: Amazon Managed Prometheus (AMP) and a self-hosted Grafana behind the same ALB and IdP (`cloudformation/03-observability.yaml`). Usage metrics reach AMP through an **ADOT collector that runs as a co-resident sidecar inside the gateway task** (`cloudformation/02-gateway.yaml`), not as a standalone service.

1.3 Scope — what the system is not

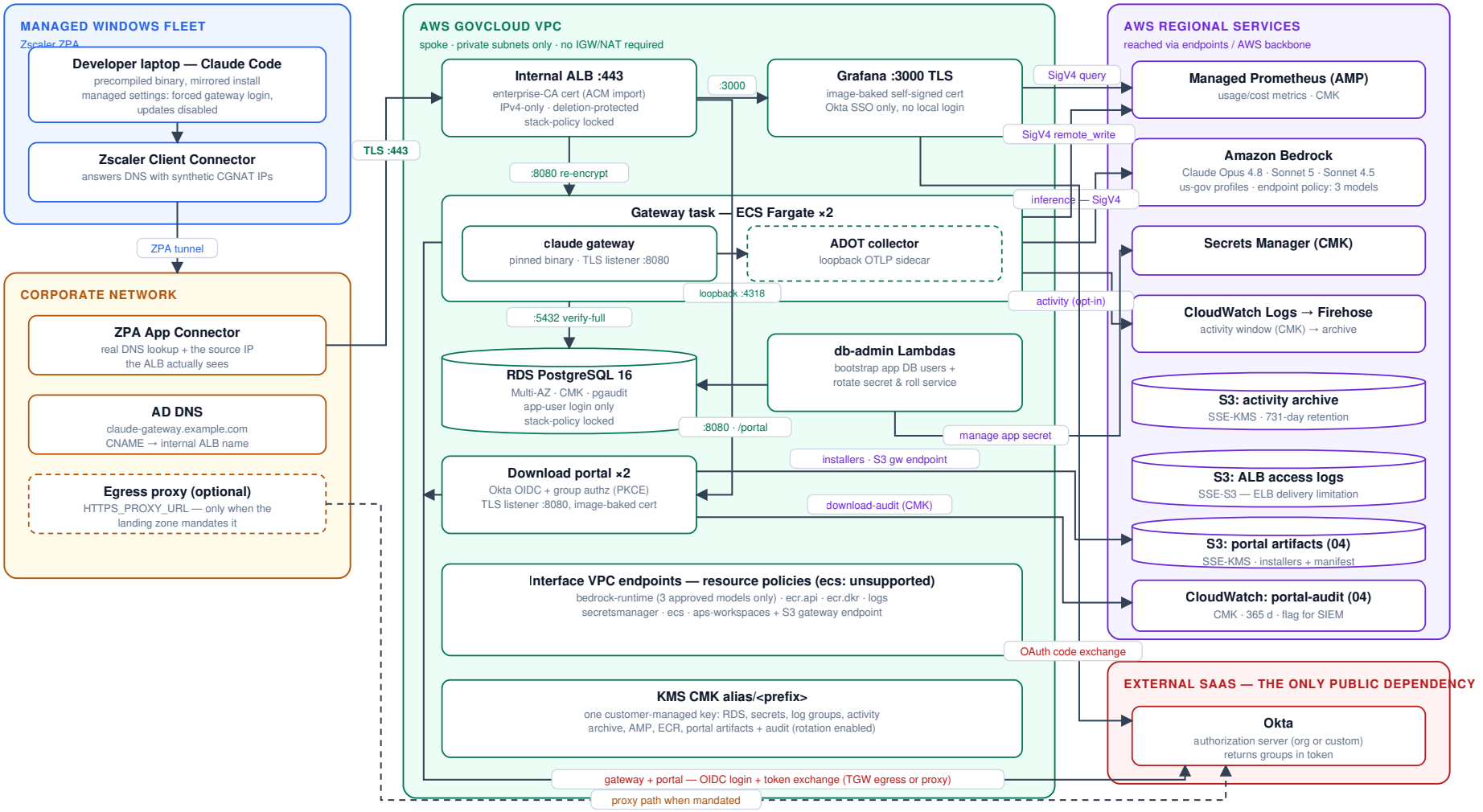
- **Not public-facing.** The ALB is internal; there is no public ingress path (`architecture.md` §1, "No public ingress").
- **Not a general AI gateway.** Exactly three Bedrock models are exposed — Opus 4.8 (the client default), Sonnet 5, and Sonnet 4.5 (serving the small/fast "haiku" role; GovCloud has no Haiku-family model) — enforced at the application, IAM, and VPC-endpoint layers (`cloudformation/02-gateway.yaml` `models:` block; `architecture.md` §10 note 5).
- **Not dependent on Anthropic-hosted infrastructure at runtime.** The client fleet never contacts Anthropic: binaries are mirrored and verified, and all auto-update paths are disabled by the user-settings `env` block the installer writes (`DISABLE_UPDATES` / `DISABLE_AUTOUPDATER`), backstopped by the mirror-only network path (`scripts/mirror/mirror-claude-release.sh`, `client/Install-ClaudeCode.ps1`).
- **Not a Node/npm distribution.** By decision, only the precompiled native `claude` binary is fielded (CLAUDE.md "User decisions").

1.4 System boundary

The authorization boundary comprises four trust zones — the managed Windows fleet (Zscaler ZPA), the corporate network (App Connectors, AD DNS, egress proxy), the AWS GovCloud workload VPC (all server components), and one external SaaS dependency, Okta. Bedrock and every other AWS service are reached over private endpoints or the AWS backbone. The single-page boundary view is reproduced below; the split views and per-hop detail are in `architecture.md` §1–§2.

System architecture & trust boundaries

Claude apps gateway in AWS GovCloud us-gov-west-1 — every arrow is TLS unless flagged



Boundary facts: no public ingress (internal ALB behind ZPA); no public egress from the inference path (Bedrock endpoint, 3-model policy); clients never contact Anthropic (mirrored, verified binaries; updates disabled).

2. System overview

At a glance (full detail in `architecture.md` §1–§2):

- Developers on managed Windows laptops run `claude`, which reaches an **internal ALB** over TLS via **Zscaler ZPA**. The ALB re-encrypts to the **ECS Fargate gateway** tasks over a TLS leaf baked into the gateway image at build time (not generated per task — see `architecture.md` §6), which authenticate the user against **Okta OIDC** and proxy inference to **Bedrock** over a private interface endpoint whose policy admits only the three approved models.
- **Session and spend state** lives in **RDS PostgreSQL 16**; the gateway connects as a least-privilege application role, never the RDS master user.
- **Usage telemetry** (tokens, cost, model, and stamped user identity) flows from the gateway to a **co-resident ADOT collector sidecar** (localhost, inside the gateway task), which remote-writes into **AMP**; it is visualized in a self-hosted **Grafana** that also authenticates through Okta.
- An optional **installer download portal** — a small ECS Fargate service on the same ALB (path `/portal`) — lets developers self-serve a pre-configured installer ZIP after Okta login and group check, with every download and denial written to a dedicated audit log (`cloudformation/04-download-portal.yaml`; §5.1, §5.4).
- Everything at rest is encrypted with a single **customer-managed KMS key** (the one documented exception is the ALB access-logs bucket, which uses SSE-S3 because ELB log delivery does not support KMS — `architecture.md` §9).

The deployment is three CloudFormation stacks (`01` database, `02` gateway, `03` observability) plus an optional fourth (`04` download portal), five container images, and a `deploy.env`-driven set of scripts. Stack dependencies and deploy order are in `architecture.md` §8.

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3. Users and roles

The system has four operational actor classes. None of the day-to-day roles holds the RDS master credential.

3.1 End-user developers

- Run `claude` on managed, Zscaler-secured Windows laptops via a **no-admin**, entirely user-scope install to the per-user profile (`client/Install-ClaudeCode.ps1`; the installer refuses a SYSTEM-context run and writes no machine/policy settings — only a user-settings `env` block).
- Reach the gateway login through the **required managed setting** (`forceLoginMethod: "gateway"` + `forceLoginGatewayUrl`) — Claude Code offers no user-selectable gateway option, so this setting must be delivered by GPO/MDM (or self-served with local admin) before the developer can sign in (`client-config.md`, `ad-request-email.md`). With it in place, `claude` auto-drives to the locked gateway login; the developer completes a **one-time Okta SSO** and receives a gateway session (`cloudformation/02-gateway.yaml` `oidc:` and `session:` blocks).
- Are gated at sign-in by **allowed email domain** (`allowed_email_domains` in `cloudformation/02-gateway.yaml`, from the `ALLOWED_EMAIL_DOMAINS` parameter). A `managed.policies` block is **always rendered**, carrying the **client model allowlist** (`availableModels`), the web/MCP tool denies, the **minimum client version floor** (`requiredMinimumVersion`, defaulting to the gateway's own version), and the small/fast-model and update-lockdown env overrides, scoped to all users via a catch-all policy that needs no group claim — client-configuration controls, not access control. Okta **group** claims are

not used as gateway access control: the `groups` scope is requested unconditionally, but for **spend caps** (per-group cost limits resolve against the groups claim). Reviewers should read gateway authorization as "authenticated Okta user in an approved email domain," with per-group policy as an available extension.
- Obtain the installer either from an operator/MDM push or **self-service from the download portal** (when stack `04` is deployed): they browse to `/portal` on the gateway FQDN, authenticate through Okta, pick their **Cost Center** and then a **Team belonging to it** (a configured cost-center→teams mapping drives the chained dropdowns; a no-script browser falls back to the equivalent two-step page flow), and receive a single ZIP whose generated `install.cmd` bakes those choices in (§5.1; `cloudformation/04-download-portal.yaml`; `docker/portal/`).

3.2 Platform operators

- Deploy and maintain the system with the `deploy.env`-driven scripts in `scripts/` (fixed order: cert → `01` → build four images → `02` → DNS/Zscaler → verify → `03` → Grafana secret → re-run `02`; `docs/operations/greenfield-deployment.md`).
- Operate from a host with Docker, AWS credentials, and egress; an **in-VPC admin/build host** is admitted to the interface-endpoint security group when `ADMIN_CLIENT_SG_ID` is set, so its AWS API calls are not black-holed by the endpoints' private DNS (`network-access-controls.md` §1).
- Never inject the RDS master secret into any task; the deploy tooling writes secrets via mode-600 `file://` temp files, never on the command line (`.claude/rules/security.md`; `scripts/common.sh` `put_secret_and_roll`).

3.3 Security and audit consumers

- Read the **DB audit trail** (pgaudit → CloudWatch), **ALB access logs** (S3), the **installer download-audit log** (when the portal is deployed — see §5.4), and, when enabled, the **AI activity stream** (see §5.4).
- The activity stream is treated as highly sensitive (bash commands, tool inputs, file paths per user): it is **opt-in**, IAM-only, CMK-encrypted, and flagged for SIEM subscription (`.claude/rules/security.md`; `cloudformation/03-observability.yaml`; `FORWARD_ACTIVITY_LOGS` in `scripts/deploy.env.example`).

3.4 Break-glass database administrator

- The RDS master role (`gw`) is **break-glass only** — held by humans for emergencies and by the db-admin Lambda, never by a running gateway task (`architecture.md` §4; `.claude/rules/security.md`). Its RDS-managed 7-day rotation therefore affects no running workload.

3.5 Grafana / dashboard role model

Grafana authenticates against the **same Okta issuer** as the gateway, with its own client and redirect URI (`/grafana/login/generic_oauth`) and **strict Okta-group → role mapping**: membership in `grafana-admins` (the `GRAFANA_ADMIN_GROUP` parameter, default `grafana-admins`) maps to Grafana Admin, and a user in no mapped group is **denied** (`architecture.md` §3; `cloudformation/03-observability.yaml`; `docs/requests/okta-request-email.md`). The local login form is disabled; the bootstrap `admin` account is break-glass only, reachable only by redeploying with `GRAFANA_DISABLE_LOGIN_FORM=false`.

3.6 Download-portal access group

The portal (optional stack `04`) authorizes on **Okta group membership**: the `AccessGroup` parameter (`ACCESS_GROUP` in `deploy.env`, placeholder default `claude-gateway-users`) names **one or more groups** (comma-separated), and a member of **any** listed group may download the installer; everyone else is denied — and the denial is audited (§5.4). The parameter is named generically so the **same group(s) can gate the gateway** if per-group access control is enabled there (§3.1), giving one org-managed membership set for both surfaces (`cloudformation/04-download-portal.yaml`; `scripts/deploy.env.example`). Because the portal's group check is unconditional — unlike the gateway, which gates on email domain — **Okta groups-claim configuration is an operating prerequisite** for the portal (§8.2).

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4. Operational environment

The system is fielded into an **AWS Landing Zone** with the following characteristics (CLAUDE.md "Landing zone"):

- **Hub-and-spoke with Transit Gateway** (not VPC peering), **central egress**; the workload VPC is a **no-NAT spoke** in the target profile. Because there is no local NAT, the no-NAT path relies on interface/gateway VPC endpoints for AWS APIs (`CREATE_SUPPORTING_ENDPOINTS`, `CREATE_BEDROCK_ENDPOINT`, `CREATE_AMP_ENDPOINT` in `scripts/deploy.env.example`).
- **Client access via Zscaler ZPA**: developer laptops reach the internal ALB through a ZPA application segment for the gateway FQDN (TCP 443), served by App Connectors that route to the workload VPC over the Transit Gateway. TLS inspection must **not** be enabled on that segment — the client pins the gateway certificate fingerprint (`docs/requests/networking-request-email.md` §3).
- **Server-side egress prerequisite**: the gateway and Grafana containers dial the **Okta issuer** for OIDC discovery and token exchange. This traffic is server-originated (no Zscaler user identity), so the workload VPC's central egress path needs an explicit **ALLOW + SSL-inspection exemption** for the Okta issuer FQDN (`docs/requests/networking-request-email.md`, "Server-side egress"). Without it the gateway fails at OIDC discovery on boot: the exchange carries the OIDC client secret and must not transit inspection infrastructure.
- **Corporate DNS**: a single CNAME in the corporate zone points the gateway FQDN at the ALB's `internal-*.elb.amazonaws.com` name — a public record returning private IPs, so no split-horizon/private-hosted-zone is required; App Connectors must be able to resolve the corporate CNAME (`docs/requests/networking-request-email.md` §2).

Per-source-IP attribution is deliberately **not** relied upon: ZPA collapses all users to a handful of App Connector IPs, so identity is carried in the Okta → gateway-session chain, not the network layer (`architecture.md` §3).

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5. Operational scenarios

The narrative walkthroughs below are the core of this ConOps. Each traces a real flow through the fielded system and cites the file that implements it.

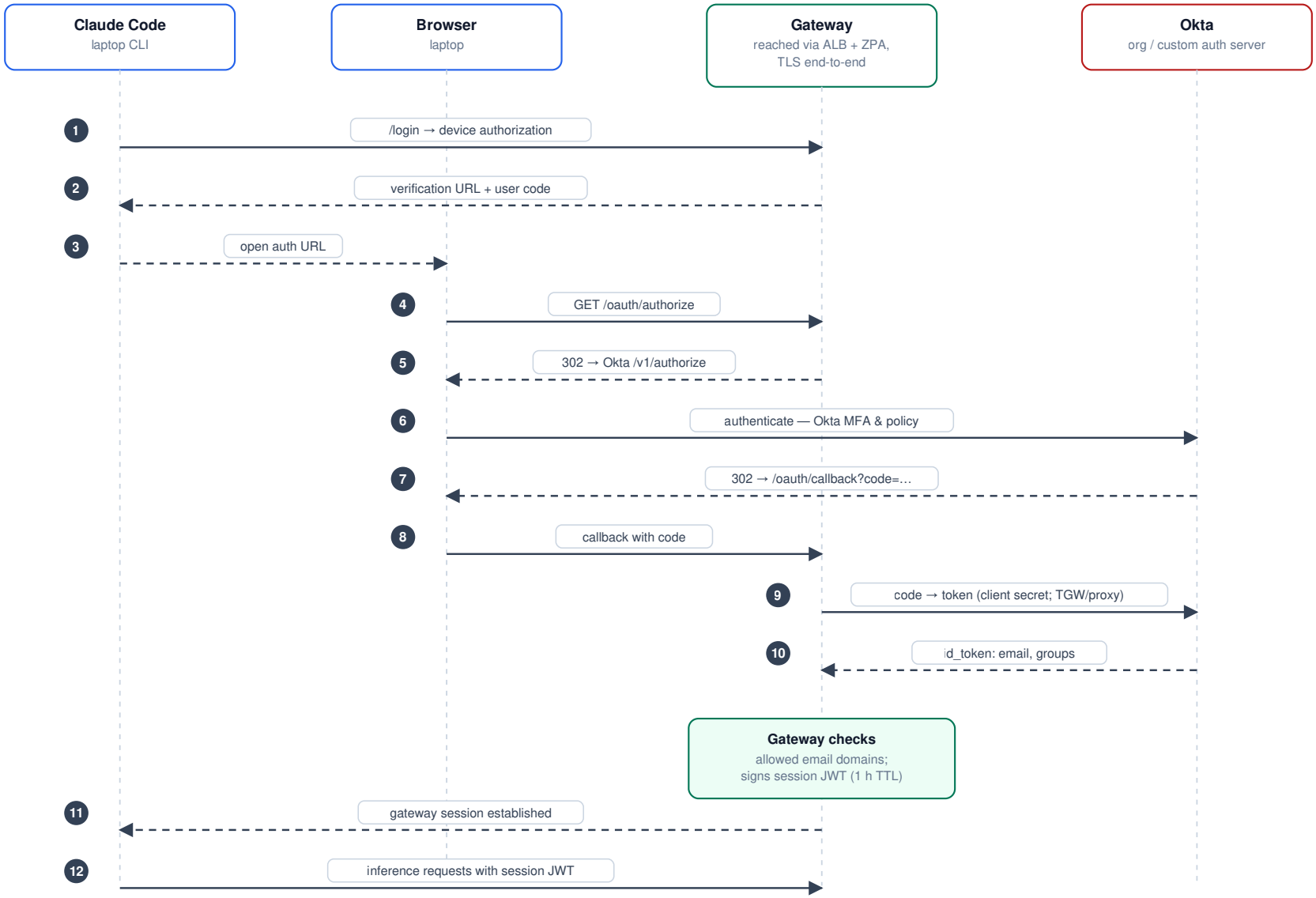
5.1 Developer onboarding and first login

1. **Offline install.** An operator has mirrored the pinned `claude` release into an internal share; the mirror **fails closed** on integrity — it refuses to proceed without GPG verification unless `ALLOW_UNVERIFIED_MANIFEST=1` is set explicitly (`scripts/mirror/mirror-claude-release.sh`; `.claude/rules/security.md`). The developer obtains the installer one of two ways:
2. **Self-service (portal, when stack `04` is deployed).** The developer browses to `https://<gateway-fqdn>/portal`, signs in through Okta (authorization-code flow with PKCE, enforced `AccessGroup` membership — §3.6), selects a **Cost Center** and then one of **its Teams** (chained dropdowns from the configured cost-center→teams mapping; two-step page flow without scripts), and downloads a single ZIP: the verified `claude.exe`, `Install-ClaudeCode.ps1`, and a generated `install.cmd` that invokes the installer with `-GatewayUrl`, `-Sha256`, `-Team`, `-CostCenter` (and update lockdown) already baked in — one double-click, no parameters to get wrong. The ZIP contents come from the portal's CMK-encrypted artifacts bucket, published from the verified mirror output by `scripts/publish-portal-release.sh` — the portal never fetches from the internet (`docker/portal/`; `cloudformation/04-download-portal.yaml`).
3. **Operator/MDM push.** The developer (or an MDM push) runs `Install-ClaudeCode.ps1` directly from the internal share.

Either way, the installer places the binary in the per-user profile **without admin rights** and writes a **user-settings `env` block** that locks down auto-updates (`DISABLE_UPDATES=1`, `DISABLE_AUTOUPDATER=1`) and stamps telemetry tags. The **gateway login itself requires a managed setting** (`forceLoginMethod: "gateway"` + `forceLoginGatewayUrl`) that this installer does **not** write — Claude Code honors those keys only from a managed source, so they are delivered by GPO/MDM (or self-served with local admin) before first login (`client/Install-ClaudeCode.ps1`; `client-config.md`, `ad-request-email.md`). After login the gateway pushes central config via `/managed/settings`. 2. **First login (Okta OIDC).** With the managed setting present the developer runs `claude`; the login screen is **locked to gateway** with the URL **pre-filled** (no picker, no URL typing — press Enter to connect), and the developer is taken through a one-time Okta authorization-code flow (PKCE, org authorization server). On success the gateway issues a session JWT that carries the Okta identity. The full sequence is diagrammed in `architecture.md` §3.

Developer authentication — Okta OIDC via the gateway

managed settings force gateway login; no local API keys exist



Audit identity chain: Okta identity → session JWT → user.id / user.email / user.groups stamped on every request and telemetry export. Grafana repeats this flow with its own client + strict group→role mapping.

1. **Certificate fingerprint trust.** Claude Code validates the ALB certificate chain, then **pins the SHA-256 fingerprint of the leaf on first connect** (trust-on-first-use, per hostname). Operators publish the expected fingerprint (printed by `import-enterprise-cert.sh`) so the developer confirms it at first login. This is exactly why TLS inspection must not sit in front of the gateway FQDN (`docs/requests/networking-request-email.md` §3; `docs/operations/greenfield-deployment.md`).

5.2 Normal use — a request

Steady-state request path (`architecture.md` §2, hop table):

Laptop `claude` → **ZPA** (TCP 443, no inspection) → **internal ALB** (TLS, FIPS policy) → **gateway task** (ALB re-encrypts to the image-baked TLS leaf on 8080 — the same key on every task running that image, not a per-task generation) → **Bedrock** (TLS + SigV4 over the interface endpoint, three approved models only). Session and spend state is read and written in **RDS** over `verify-full` TLS, with the gateway authenticating as the least-privilege application role (`gateway_app` / `gateway_app_clone`), never the master user (`architecture.md` §4). WebSearch is disabled on gateway sessions, so the inference path has no public egress (`architecture.md` §1).

5.3 Usage and cost monitoring

Each developer's user-settings `env` block can stamp **OTEL resource attributes** — `cost_center` and `team` — via `OTEL_RESOURCE_ATTRIBUTES`, set by the installer's `-CostCenter` / `-Team` parameters (`client/Install-ClaudeCode.ps1`). The gateway additionally stamps **identity** (`user.id` / `user.email` / `user.groups` from the Okta session) onto every telemetry export (`architecture.md` §3, §5). Metrics flow gateway → **co-resident ADOT collector sidecar** → **AMP** (SigV4 remote-write) → **Grafana**, where operators authenticate through Okta SSO and read the provisioned usage/cost dashboard (`cloudformation/02-gateway.yaml`, `cloudformation/03-observability.yaml`; `architecture.md` §5). Metrics carry 150-day AMP retention.

The collector is not a network service: it runs as a **sidecar container in the gateway task**, sharing that task's boundary and network namespace, and the gateway forwards to it over **loopback** (`http://localhost:4318`) — there is no over-the-wire telemetry hop. The sidecar remote-writes to AMP and to the activity log group under the **gateway task role** (scoped to this workspace and this log group only), and its OTLP listener binds `127.0.0.1` so nothing off the task can reach it. SC-8 is met by absence of network transmission — a stronger posture than an internal-CA TLS listener, and with no new PKI to custody (SC-17/SC-12). The design is also the only one the gateway permits: it rejects a non-HTTPS telemetry forward URL unless the host is loopback (`security-assessment-2026-07.md`; `architecture.md` §10 note 1).

5.4 Audit

Three independent audit surfaces, at increasing sensitivity (`architecture.md` §5 data-flow table):

- **DB audit (pgaudit).** DDL, role changes, and writes (no bind values) are exported from RDS to a CMK-encrypted CloudWatch log group (`cloudformation/01-database.yaml`).

- **ALB access logs.** Source connector IPs, URIs, and timings land in an S3 bucket (SSE-S3, IAM-only, lifecycle expiry) (`cloudformation/02-gateway.yaml`; `architecture.md` §9 note).
- **Installer download audit (portal, when deployed).** Every portal download **and every denial** writes one JSON record — timestamp, verified Okta identity (`user_email`, `user_groups`), selected team and cost center, release version, binary SHA-256, ALB-attested source IP, outcome — to a **dedicated CMK-encrypted CloudWatch log group** (retention `PORTAL_AUDIT_RETENTION_DAYS`, default 365; flag for SIEM). It is deliberately separate from, and never routed into, the activity stream below (`docker/portal/`; `cloudformation/04-download-portal.yaml`).
- **AI activity stream (opt-in, highly sensitive).** When `FORWARD_ACTIVITY_LOGS=true`, bash commands, tool inputs, and file paths per user are streamed to a CMK-encrypted CloudWatch window and archived to S3 (SSE-KMS), IAM-only and flagged for SIEM subscription (`cloudformation/03-observability.yaml`; `scripts/deploy.env.example`; `.claude/rules/security.md`). Prompt content is redacted. This stream's access surface is deliberately never widened.

5.5 Administration — deploy, update, and rotation

- **Deploy / update flow.** The load-bearing order is cert → `01` database → build all four images → `02` gateway → DNS/Zscaler → `verify-gateway.sh` → `03` observability → set Grafana Okta secret → re-run `02` (`docs/operations/greenfield-deployment.md`; CLAUDE.md "Deploy model"). Scripts persist their outputs back into `deploy.env` (`set_env_var` in `scripts/common.sh`), so there are no copy-paste steps. Container image tags are immutable — an image change means a new tag and a service roll (`.claude/rules/scripts.md`).
- **Automated DB-credential rotation.** The application DB secret (`<prefix>/db-app-user`) rotates on a schedule (default 90 days, `APP_SECRET_ROTATION_DAYS`) using an **alternating-user** scheme — the db-admin Lambda flips between `gateway_app` and `gateway_app_clone` and **forces the gateway service to roll** as part of `finishSecret` (`docker/db-admin/app.py`; `architecture.md` §4). The previous credential stays valid until the next rotation. The first rotation fires at stack creation as an automatic end-to-end validation (asynchronous — see §8).
- **Manual secret rotation.** Okta client secrets (gateway, Grafana, and — when deployed — the download portal) are set out-of-band via `set-okta-secret.sh` / `set-grafana-oidc-secret.sh` / `set-portal-oidc-secret.sh`, which prompt hidden, write via mode-600 `file://`, and roll the consuming service (`scripts/common.sh put_secret_and_roll`). The JWT session-signing secret has a documented prepend → roll → remove runbook (`architecture.md` §6).
- **Portal operations (optional stack `04`).** Deployed after `02` (independent of `03`): build/push the portal image (`build-and-push-portal.sh`), deploy `deploy-download-portal.sh`, set the portal's Okta secret, then publish the pinned release to the artifacts bucket with `publish-portal-release.sh <version>` — which reuses the GPG-verified mirror output. Serving a **new client version** to developers is that one publish command plus updating `PORTAL_RELEASE_VERSION` (`docs/operations/om-runbooks.md` §5; `docs/operations/greenfield-deployment.md`).

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6. Modes of operation

6.1 Normal

All three stacks healthy: developers log in through Okta, inference is served from Bedrock, telemetry flows to Grafana, and audit surfaces are recording.

6.2 Degraded

- **Observability stack (03) down → gateway unaffected.** Telemetry forwarding is optional and gated on the observability stack existing; the deploy-gateway guard only disables forwarding on a definitively-missing stack, and a gateway with no OTLP target continues to serve inference. Usage dashboards go stale; developer service does not (`cloudformation/02-gateway.yaml` telemetry toggle).
- **Okta unreachable → no new logins, existing sessions continue.** OIDC is the authentication front door; if the issuer is unreachable (or the server-side egress ALLOW/exemption is missing), **new** logins fail closed, but sessions already holding a valid JWT keep working until their TTL expires (`SESSION_TTL_HOURS`; `cloudformation/02-gateway.yaml` `session:` block). The same failure occurs at gateway boot if the server-side egress ALLOW / SSL-inspection exemption for the issuer FQDN is withdrawn (§4).
- **Collector shares the gateway task lifecycle — and fails CLOSED by default (AU-5).** The ADOT collector is a sidecar in the gateway task, not a separate service, so there is no independent collector redeploy: a gateway roll cycles the collector with it. With the default `TelemetryFailClosed=true`, the sidecar is Essential and health-checked and the gateway container waits on it: a transient collector crash restarts in place (container restart policy), but a collector that stays failed or hangs unhealthy **stops the task** — the gateway does not serve traffic while telemetry/audit processing is down; ECS replaces the task and the ALB routes to the peer task meanwhile. Orgs that prefer availability over auditability set `TELEMETRY_FAIL_CLOSED=false` (telemetry may then gap silently while inference keeps serving) and record that deviation in the SSP. On every task stop — including secret-rotation service rolls — stop ordering is gateway-first, collector-last with a 120 s drain, so the final telemetry buffer gets a best-effort flush to AMP/CloudWatch rather than being dropped on the spot (delivery is not guaranteed — a failing AMP endpoint at shutdown can still lose the last batch). The **end-to-end control is the missing-telemetry alarm** (observability stack): container health proves the collector is alive, while the alarm proves the AMP remote-write **pipeline** is alive — it fires after `MISSING_TELEMETRY_ALARM_MINUTES` (default 15) of total ingestion silence on the workspace, covering IAM/endpoint breakage and a full AMP-side outage that no health check can see. A continuous heartbeat — the sidecar scrapes its own `otelcol_*` self-metrics over loopback into the same remote-write pipeline — keeps that alarm meaningful on an idle fleet, where the bursty client metrics alone would otherwise look like a broken pipeline. Because the heartbeat shares the same pipeline as client metrics, the alarm cannot by itself prove that **client** usage metrics specifically are landing: a translation-layer failure that silently drops only `claude_code_*` points (the delta-temporality incident of 2026-07-24, since fixed at the client) leaves total ingestion above zero and this alarm stays OK. Detecting that narrower failure class today is a manual diagnostic — `scripts/diagnostics/amp-query.py`, or watching the collector's own `otelcol_exporter_prometheusremotewrite_failed_translations` metric — not an alarm (`docs/operations/troubleshooting.md`; O&M runbook 9 triage).
- **Download portal down → no self-service installs, gateway unaffected.** The portal is an optional, separate stack behind its own `/portal` listener rule; its failure (or absence) does not touch the inference path. Installers remain available from the internal share via the operator/MDM route (§5.1).

6.3 Maintenance

- **Image updates.** Rebuild and push the affected image under a **new immutable tag**, then roll the service; always push rebuilt images *before* a stack update that changes what the task definition expects (`.claude/rules/scripts.md`).
- **Certificate rotation.** The enterprise leaf is imported into ACM and does **not** auto-renew; a `DaysToExpiry` alarm fires at 30 days, and re-import re-triggers a one-time client fingerprint-trust prompt, so operators coordinate a heads-up before each renewal (`architecture.md` §6; `docs/requests/networking-request-email.md` §1).

- **Teardown / replacement discipline.** The ALB and RDS instance are protected three ways (deletion protection, fixed names, stack policy denying replace/delete); several encryption-at-rest choices (RDS CMK, AMP CMK) are **day-one decisions** that cannot be changed by an in-place update (`architecture.md` §8; `.claude/rules/cloudformation.md`).

Detailed operations-and-maintenance procedures are collected in the O&M runbooks document, `docs/operations/om-runbooks.md`.

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7. Security posture summary and accepted risks

Stated up front, per the review-package convention. The source of truth for finding-by-finding status is `security-assessment-2026-07.md`; the accepted-risk register is `architecture.md` §10.

The implemented posture: a single customer-managed KMS key encrypts everything at rest (bar the ALB-logs bucket, an ELB platform limitation); no plaintext network hop anywhere — TLS in transit on every hop, and the telemetry leg is a loopback sidecar with no network transmission at all (below); RDS `verify-full`; pgaudit; VPC-endpoint and IAM policies scoped to the three approved models and this account's exact ARNs; explicit (no default-allow) security-group egress; and a least-privilege application DB user with self-rolling rotation.

Accepted / deferred risks a reviewer should see immediately:

- **There is no plaintext telemetry leg to accept.** Earlier review packages recorded the gateway→collector OTLP hop as an accepted plaintext-but-SG-scoped risk. That acceptance is **withdrawn**: the gateway refuses a non-HTTPS telemetry forward URL unless the host is localhost, so the hop was never implementable, and the collector runs as a **co-resident localhost sidecar** in the gateway task instead (§5.3). SC-8 is met by absence of network transmission, with no new PKI to manage (`architecture.md` §10 note 1). Surfaced here rather than silently dropped because it reverses a decision recorded in earlier packages.
- **S3 Object Lock / WORM on audit buckets is deferred**, by decision. The archives are CMK-encrypted, IAM-scoped, and `Retain`-protected, but a sufficiently privileged principal could still delete or shorten retention — and that exposure is not uniformly narrowed by versioning today: only the portal artifacts bucket (04) has `VersioningConfiguration` enabled; the activity-archive and Bedrock prompt-log buckets (03) do not. Revisit if AU-9 WORM retention is mandated (`architecture.md` §10 note 3).
- **Spend enforcement fails closed.** `enforcement.fail_closed_on_error: true` means a spend-store (RDS) outage blocks inference **fleet-wide** rather than allowing uncapped requests — a deliberate availability-for-control trade. Recovery procedure: `docs/operations/cost-controls.md` §5.
- **ALB access-logs bucket uses SSE-S3, not the CMK** — an AWS platform limitation (ELB log delivery does not support KMS), not an oversight; the bucket blocks public access and expires on a lifecycle (`architecture.md` §9, §10 note 2).
- `ClientIngressCidr` **is the pre-auth reachability bound** and must be set per deployment to the ZPA App Connector source range: the same ALB security group fronts the gateway, `/grafana`, and the portal (all three of which independently require Okta with group or domain checks).

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8. Assumptions, constraints, and dependencies

8.1 Assumptions and constraints

- **GovCloud model availability.** The three configured models are **Opus 4.8** (`us-gov.anthropic.claude-opus-4-8`), **Sonnet 5** (`us-gov.anthropic.claude-sonnet-5`), and **Sonnet 4.5** (`us-gov.anthropic.claude-sonnet-4-5-20250929-v1:0`) in `us-gov-west-1`. Sonnet 5 is the configured Sonnet-tier default; its inference-profile ID is un-dated, like Opus 4.8. Sonnet 4.6 was never offered in GovCloud, and no Haiku-family model is (Sonnet 4.5 fills the small/fast role). Model IDs must always be verified against the Bedrock console before changing defaults (`scripts/deploy.env.example`; CLAUDE.md "GovCloud model availability").
- **Template, not a deployment.** No organization-specific value is hardcoded; each is a CloudFormation parameter or a `deploy.env` variable (`.claude/rules/security.md`).
- **Client distribution is the precompiled native binary only**; Grafana auth is Okta SSO; Object Lock is deferred (CLAUDE.md "User decisions").
- **Account-level CloudTrail is assumed, not provisioned by this repo.** No template here creates an `AWS::CloudTrail::Trail`; management-event coverage (AU-2/AU-6) is assumed inherited from the landing zone's account baseline, not verified by this project. Object-level (S3 data-event) logging on the activity-archive and Bedrock prompt-log buckets is not configured anywhere in these templates; if data-event coverage on those two buckets is required for the ATO package, it must be added to the org's trail (or a dedicated trail provisioned) as a separate action.

8.2 Organizational dependencies (prerequisites)

These must be provided by the operating organization before the system can operate; templates for the requests exist in the repo:

- **Enterprise-CA TLS certificate** for the gateway FQDN (serverAuth EKU), imported into ACM (`docs/requests/networking-request-email.md` §1; `scripts/import-enterprise-cert.sh`).
- **Internal DNS CNAME** for the gateway FQDN at the ALB name (`docs/requests/networking-request-email.md` §2).
- **Zscaler policy**: a ZPA app segment (or ZIA bypass) for the gateway FQDN with no TLS inspection, **and** the server-side egress ALLOW + SSL-inspection exemption for the Okta issuer FQDN (`docs/requests/networking-request-email.md` §3).
- **Okta OIDC application** — a confidential Web app on the org authorization server, all redirect URIs registered (gateway, Grafana, and — if the portal is deployed — `/portal/oauth/callback`), groups returned, and the admin group provisioned (`docs/requests/okta-request-email.md`).
- **Okta groups claim configured — hard prerequisite for the portal.** The `groups` *scope* alone yields group membership in **neither** the ID token nor userinfo on an org authorization server; an Okta admin must configure a groups **claim**, or the portal denies every user. The gateway gates on email domain and Grafana is optional, so the portal is the surface that makes this a hard prerequisite (`docs/requests/okta-request-email.md`; `docs/operations/om-runbooks.md`).

8.3 Operational status

The deployment has completed end-to-end validation in the target GovCloud account and operates in steady state: developer sign-in through Okta, inference against the three approved Bedrock models, the least-privilege DB user and its automated rotation, usage telemetry reaching AMP through the loopback sidecar, Grafana dashboards, the audit surfaces of §5.4, and the download portal.

Re-deployments into a new account follow `docs/operations/greenfield-deployment.md`, whose validation checklist is the gate before a deployment is treated as fielded; steady-state procedures are in `docs/operations/om-runbooks.md` and failure-mode triage in `docs/operations/troubleshooting.md`.